

Learning with Calibration: Exploring Test-Time Computing of Spatio-Temporal Forecasting —— NeurIPS 2025 (Spotlight)

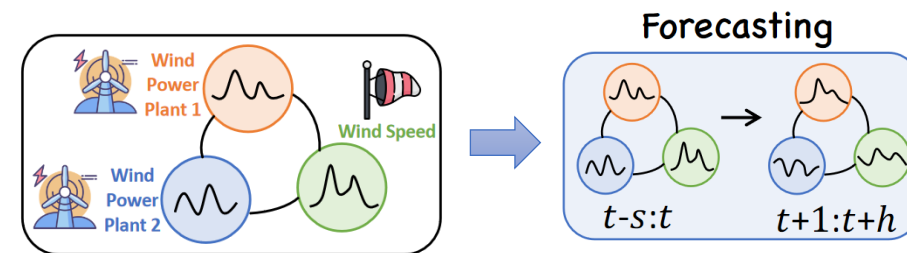
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Definition and application of STF

- What is Spatio-Temporal Forecasting (STF) ?
 - It can be considered as a complex extended task of multivariate time series forecasting ($|Location|=1$), where different variables have spatially dependent properties.



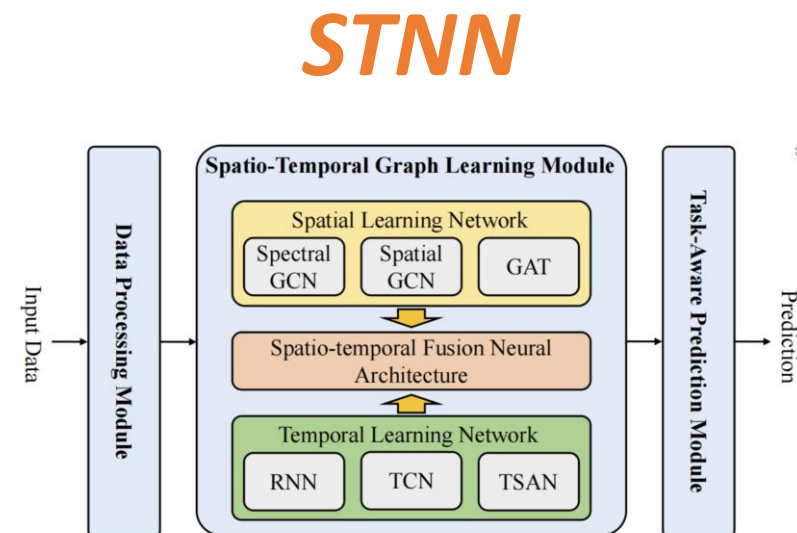
Jin, Ming, et al. TPAMI 2024

- Application of Spatio-Temporal Forecasting
 - Traffic Flow Optimization
 - Air Quality Management
 - Win Energy Allocation



Neural Components of modeling STF tasks

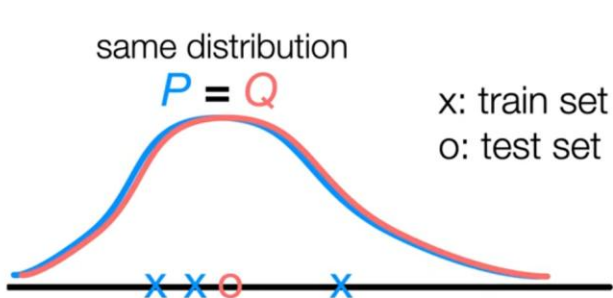
- How to modeling STF task from architecture level ?
 - Design by different relational inductive biases
 - Spatial Modeling: Graph Operator
 - Temporal Modeling: Sequence Operator
 - Fusion them to focus on spatio-temporal correlations



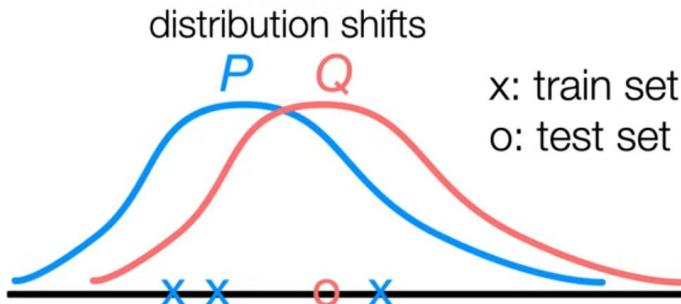
Jin, Guangyin, et al. TKDE 2023



Spatio-temporal Forecasting in Real-World



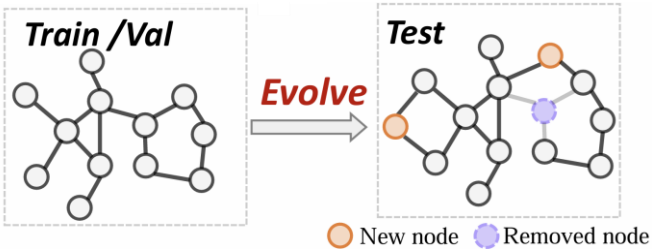
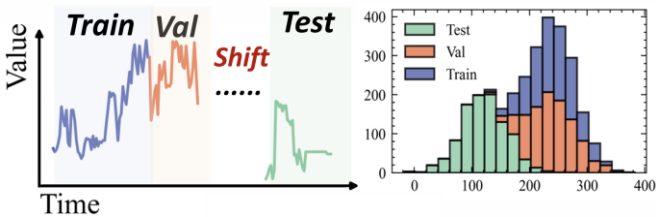
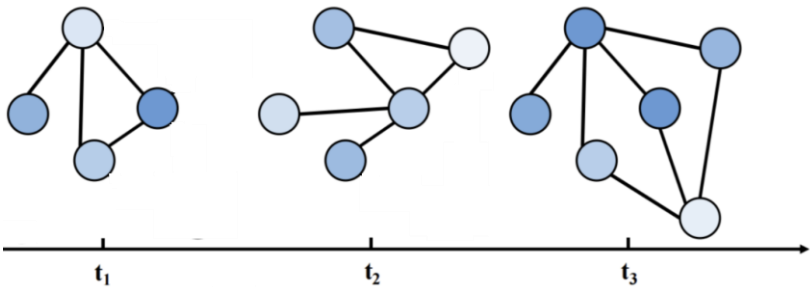
- In theory: same distribution for training and testing



- In the real word: distribution shifts are everywhere

- Spatio-Temporal Forecasting in Real-World ?

- Dynamic! & Streaming!



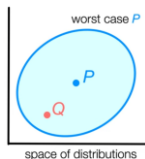
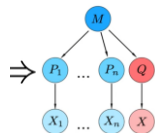
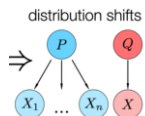


From Learning View to deal with dynamic change

- How to dealing with it from learning level ?

- OOD Learning**

- Domain adaptation**
 - Data from the test distribution
- Domain generalization**
 - Data from the meta distribution
- Adversarial robustness**
 - Topological structure of the test distribution



- Continual Learning**

- Replay based

- Test-Time Learning**

Table 1 Formal comparison of different spatio-temporal learning paradigms for generalization from the perspective of data and learning. s denotes the source domain, t denotes the target domain, x and y denote the samples and labels sampled from $\mathbf{X}$ and $\mathbf{Y}$, respectively. OOD learning expects inputs sampled from any environment $e^* \sim \mathcal{E}$ to be valid, while others are only optimized for the current training or test environment e . In particular, continual fine-tuning divides the target domain into multiple stages and optimizes for a specific stage τ environment e^τ . $\times$ means not involved.

Setting	Example Works	Data Perspective		Learning Perspective	
		Source	Target	Train-Time	Test-Time
OOD Learning	STONE (Wang et al., 2024a), CoST (Xia et al., 2023)	$(\mathbf{X}^*, \mathbf{Y}^*)$	$\times$	$\min_{f_\theta} \max_{e^* \in \mathcal{E}} \mathbb{E}_{(x,y) \sim P(\mathbf{X}^*, \mathbf{Y}^*, e^*)} [L(f_\theta(x), y)]$	$\times$
Continual Fine-Tuning	EAC (Chen and Liang, 2025), TrafficStream (Chen et al., 2021)	$\times$	$(\mathbf{X}^t, \mathbf{Y}^t)$	$\min_{f_{\theta^\tau}} \mathbb{E}_{(x,y) \sim P(\mathbf{X}^t, \mathbf{Y}^t, e^\tau)} [L(f_{\theta^\tau}(x), y)]$	$\times$
Test-Time Training	TTT-ST (Chen et al., 2024)	$\mathbf{X}^s$	$\mathbf{X}^t$	$\min_{f_\theta} \mathbb{E}_{(x,y) \sim P(\mathbf{X}^s, \mathbf{Y}^s)} [L(f_\theta(\tilde{x}), x)]$	$\min_{f_\theta} \mathbb{E}_{(x,y) \sim P(\mathbf{X}^t, \mathbf{Y}^t)} [L(f_\theta(\tilde{x}), x)]$
Online Continual Learning	DOST (Wang et al., 2024c)	$\times$	$\mathbf{X}^t$	$\times$	$\min_{f_{\theta(\delta)}} \mathbb{E}_{(x,y) \sim P(\mathbf{X}^t, \mathbf{Y}^t)} [L(f_{\theta(\delta)}(x), y)]$
Test-Time Computing	ST-TTC (Our)	$\times$	$\mathbf{X}^t$	$\times$	$\min_{g_\theta} \mathbb{E}_{(x,y) \sim P(\mathbf{X}^t, \mathbf{Y}^t)} [L(g_\theta(f_\theta(x)), y)]$

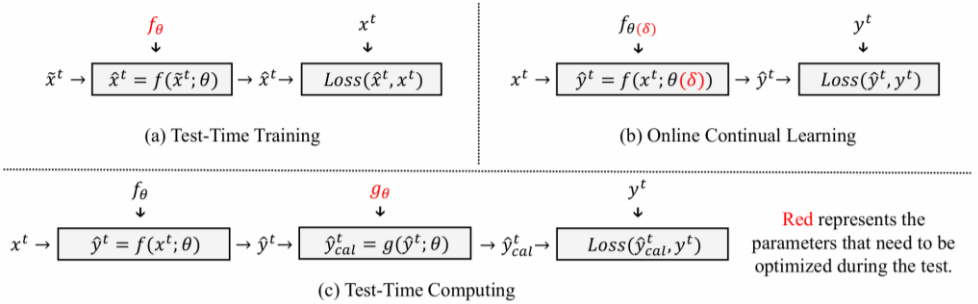
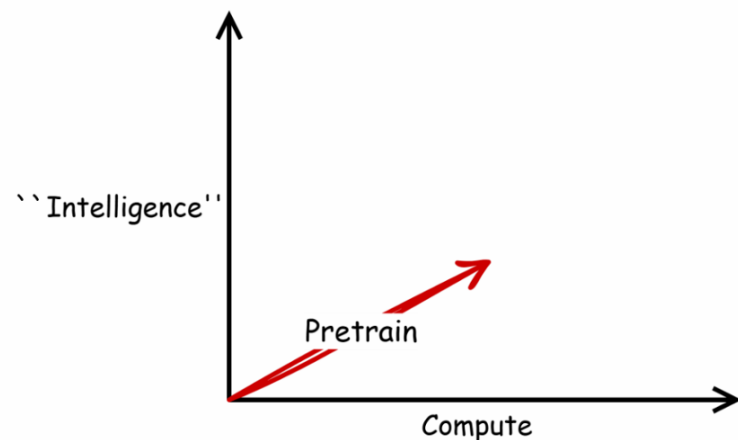


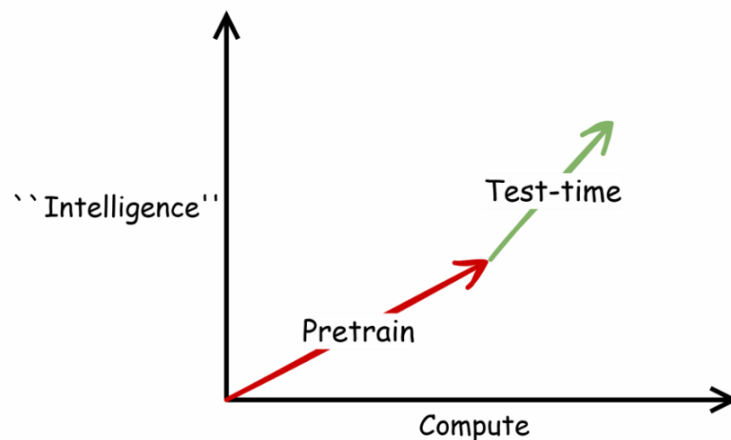
Figure 1 Conceptual visualization comparison of different spatio-temporal learning paradigms under test environment. (a) *Test-Time Training* requires the use of additional pretext tasks in the training and test phases to optimize the self-supervision head or the overall model parameters f_θ . (b) *Online Continual Learning*, by optimizing some internal parameters $f_{\theta(\delta)}$ of the model, requires additional modifications to the internal architecture of the network. Our (c) *Test-Time Computing* method only requires a lightweight calibrator g_θ , which is a seamless and lightweight plug-and-play module.



Hype phenomena in test-time learning

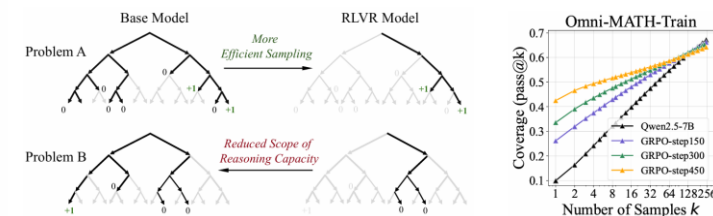


Increasing pretraining-time compute yields consistent performance improvements



Increasing test-time compute yields consistent performance improvements

- What to Scale
- **Parallel Scaling**
Generating multiple outputs in parallel.
 - **Sequential Scaling**
Updating intermediate states iteratively.
 - **Hybrid Scaling**
Balancing exploration and exploitation.
 - **Internal Scaling**
Determining autonomously how much computation to allocate for reasoning.



Yue, Yang, et al. NeurIPS 2025.

Test-time scaling (also aka **inference scaling**, **test-time compute**) progressively elicits the model's intelligence in the test-time phase.

Does RL help with generalization or sharp distribution?

There's still debate.



A History of Test-Time Learning

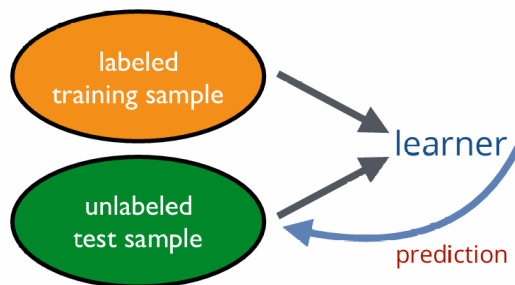
“Transductive Learning” (Gammerman et al., 1998) in the 1990s

Induction vs Transduction

- Inductive scenario:



- Transductive scenario (Vapnik, 1998):



Generalization Bounds

- VC-dimension bounds (Vapnik, 1998; Cortes and MM 2007).
- PAC-Baeyesian bounds (Derbeko, El-Yaniv, and Meir, 2004).
- Stability bounds (El-Raniv and Pechyony 2008; Cortes, MM, Pechyony, Rastogi, 2008 and 2009).
- Rademacher complexity bounds (El-Raniv and Pechyony 2007).

The rise of test-time adaptation in CV domain

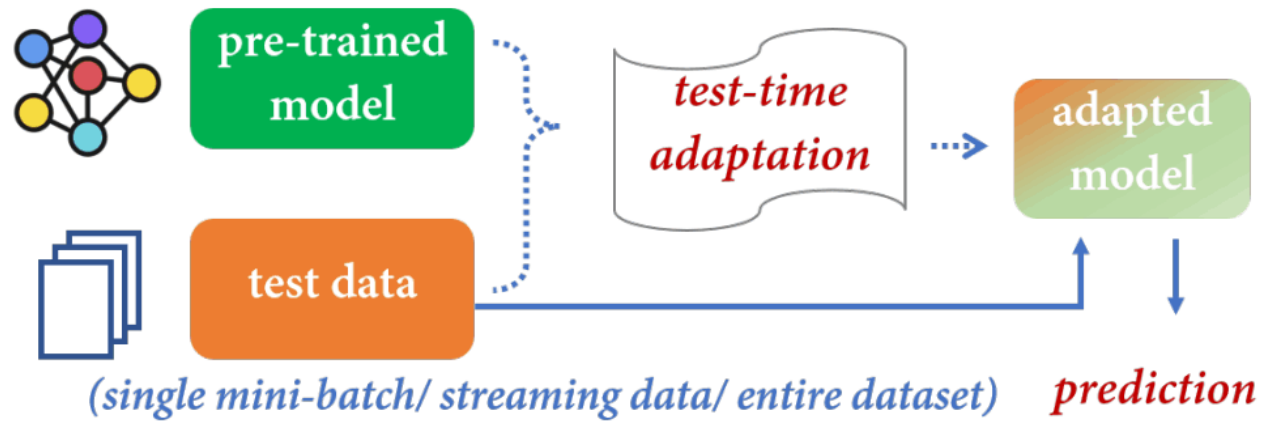


Fig. 1. The test-time adaptation (TTA) paradigm aims to adapt the pre-trained model to various types of unlabeled test data, including single mini-batch, streaming data, or an entire dataset, before making predictions. During the adaptation process, either the model or the input data can be altered to improve performance.

training

$$\min_{\theta_e, \theta_s, \theta_m} \mathbb{E}_P \left[\begin{aligned} &\ell_m(x, y; \theta_e, \theta_m) \\ &+ \ell_s(x, y_s; \theta_e, \theta_s) \end{aligned} \right]$$

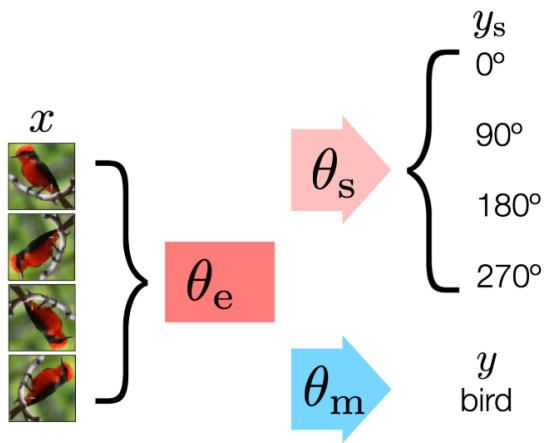
testing

$$\min_{\theta_e, \theta_s} \mathbb{E}_Q \left[\ell_s(x, y_s; \theta_e, \theta_s) \right]$$

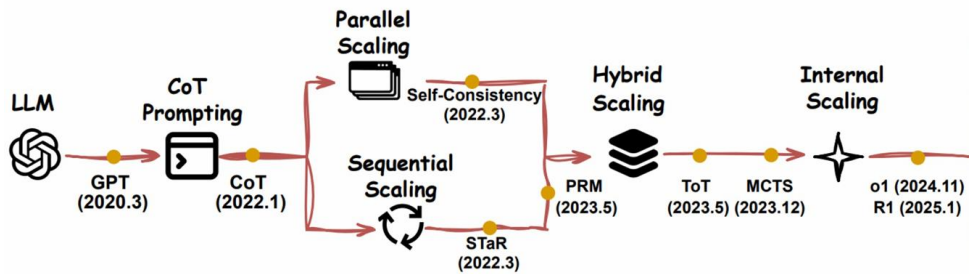
$\rightarrow \theta(x)$: make prediction on x



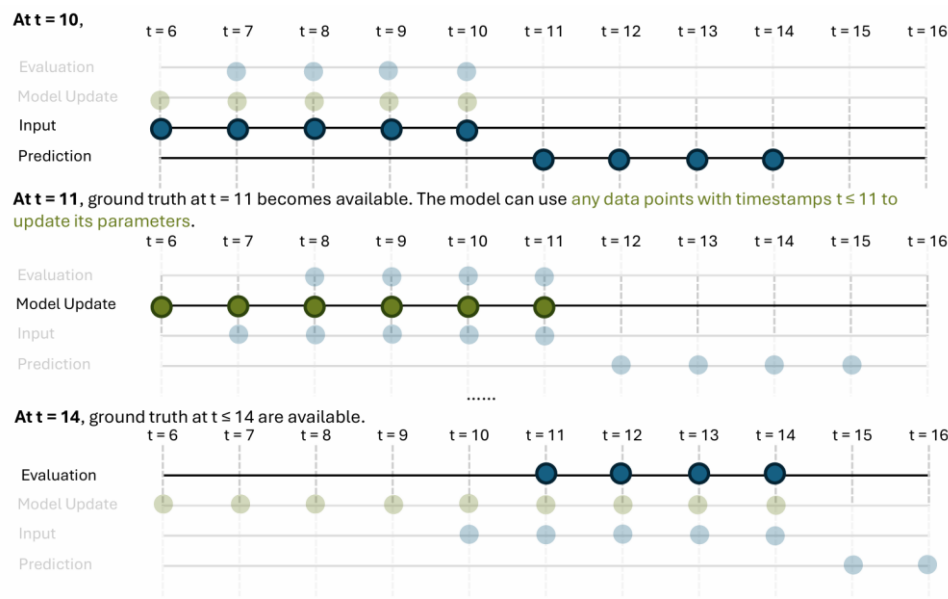
What's different between language / vision and ST data



No Label ? Self-Supervision



No Label ? Self-Supervision



Can access Label ? Supervision



Our Solution: ST-TTC

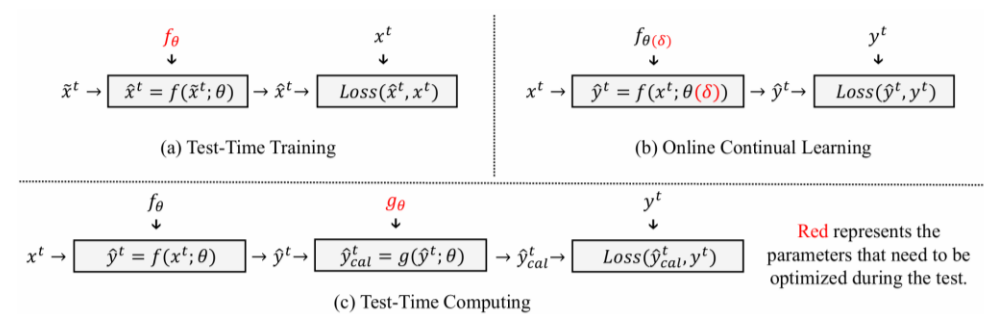


Figure 1 Conceptual visualization comparison of different spatio-temporal learning paradigms under test environment. (a) *Test-Time Training* requires the use of additional pretext tasks in the training and test phases to optimize the self-supervision head or the overall model parameters f_θ . (b) *Online Continual Learning*, by optimizing some internal parameters $f_{\theta(\delta)}$ of the model, requires additional modifications to the internal architecture of the network. Our (c) *Test-Time Computing* method only requires a lightweight calibrator g_θ , which is a seamless and lightweight plug-and-play module.



What to compute ?

- **Motivation**

amplitude (e.g., increased or decreased traffic peaks due to seasonal changes) or phase shifts (e.g., peak hours are advanced or delayed due to traffic congestion)

→ how to design an effective calibrator that can efficiently capture such gradual systematic bias from the pattern to correct the prediction errors caused by non-stationarity, while avoiding overfitting to random noise?

- **Key Challenges**



1 the degree of non-stationarity varies across spatial nodes

2 full-spectrum parameterization is computationally expensive.



Tuning Principle I : Expand

- Empirical Observation
- Theoretical Analysis

Algorithm 1 Spectral Domain Calibrator

Require: Pre-trained backbone f_θ , Test input x , Horizon length T , Number of nodes N , Groups G

Ensure: Calibrated output $\hat{y}^{\text{cal}}$

1: Get the backbone predictions: $\hat{y} = f_\theta(x) \in \mathbb{R}^{N \times T}$

2: Compute $M \leftarrow \frac{T}{2} + 1$

▷ **I: Spatial-aware Decomposition**

3: Apply real-to-complex FFT along time dimension for each node: $Y_f \leftarrow \text{rFFT}(\hat{y}) \in \mathbb{C}^{N \times M}$

4: Decompose: $A \leftarrow |Y_f|$, $P \leftarrow \angle Y_f$

▷ **II: Group-wise Modulation**

5: **for** $g = 1, \dots, G$ **do**

6: Get group index: $start \leftarrow (g-1)\lfloor M/G \rfloor + 1$, $end \leftarrow \begin{cases} M & g = G \\ \lfloor M/G \rfloor & \text{otherwise} \end{cases}$

7: Get learnable offsets: $\lambda_g^\alpha \in \mathbb{R}^{N \times 1}$, $\lambda_g^\phi \in \mathbb{R}^{N \times 1}$

8: Modulate group-slice: $A'_g \leftarrow A[:, start : end] \odot (1 + \lambda_g^\alpha)$, $P'_g \leftarrow P[:, start : end] + \lambda_g^\phi$

9: Reconstruct slice: $Y'_f[:, start : end] \leftarrow A'_g \odot e^{j P'_g}$

10: **end for**

▷ **III: Inverse Transform**

11: Inverse FFT: $\hat{y}^{\text{cal}} \leftarrow \text{irFFT}(Y'_f) \in \mathbb{R}^{N \times T}$

12: **return** $\hat{y}^{\text{cal}}$

Tuning Principle I: Prompt Parameter Pool Can Continuously Adapt to Heterogeneity Property.



How to compute ?

- **Motivation -> Solution**

Parameter Inflation → An intuitive solution is to similarly *apply low-rank matrix approximations to the prompt parameter pool, thereby reducing the number of learnable parameters while maintaining performance.*

- **Insight**

Background: numerous well-established studies that enhance the efficiency of spatio-temporal prediction and imputation tasks using compressed sensing and matrix / tensor decomposition



These study typically focus solely on the raw ST-data, how about prompt parameter pool ?



Intuition : If the prompt parameter pool also exhibits **similar redundancy** characteristics and **remains in the continuous setting**, then we are good.



Tuning Principle II : Compress

- Empirical Observation

Algorithm 2 Flash Gradient Update Mechanism

Require: Test spatio-temporal sample stream $\{x_t\}_{t=1}^B$, Pre-trained backbone f_θ , Streaming memory queue $\mathcal{Q}$, Queue size T (equal to horizon length)

Ensure: Spectral domain calibrator g_θ , Prediction collection of test samples $\{\hat{y}_t^{cal}\}_{t=1}^B$

```
1: Initialize Calibrator module  $g_\theta = (\lambda^\alpha, \lambda^\phi)$ , empty queue  $\mathcal{Q}$ 
2: for each timestep  $t = 1, 2, \dots$  do
3:   Receive  $x_t$ , compute default prediction  $\hat{y}_t = f_\theta(x_t)$ 
4:    $\triangleright$  I: Streaming Memory Queue
5:   Use Algorithm 1 to obtain the calibration results:  $\hat{y}_t^{cal} = g_\theta(\hat{y}_t; \lambda)$ 
6:   Record ground truth:  $y_t$  (collected by the value of  $x_t$ , available  $T$  time steps in the future)
7:    $\mathcal{Q}.enqueue((x_t, y_t))$ 
8:    $\triangleright$  II: Flash Gradient Update
9:   if  $\text{len}(\mathcal{Q}) > T$  then
10:     $(x_o, y_o) = \mathcal{Q}.dequeue()$ 
11:    Use Algorithm 1 to obtain the calibration results:  $\hat{y}_o^{cal} = f_\theta(x_o)$ 
12:    Update:  $\lambda \leftarrow \lambda - \eta \nabla_\lambda L(y_o, \hat{y}_o^{cal})$ 
13:  end if
14: end for
15: return  $g_\theta, \{\hat{y}_t^{cal}\}_{t=1}^B$ 
```

Tuning Principle II: Prompt Parameter Pool Can Continuously Satisfy the Low-rank Property.



Experimental Setup

• Dataset and Evaluation Protocol

Table D.1 Summary of datasets used for our experiments. Degree: the average degree of each node. Meta: the number of metadata associated with each node. Data Points: multiplication of nodes and frames. M: million (10^6).

Source	Dataset	Nodes	Time Range	Frames	Sampling Rate	Data Points
(Song et al., 2020)	PEMS03	358	09/01/2018 – 11/30/2018	26,208	5 minutes	9.38M
	PEMS04	307	01/01/2018 – 02/28/2018	16,992	5 minutes	5.22M
	PEMS07	883	05/01/2017 – 08/06/2017	28,224	5 minutes	24.92M
	PEMS08	170	07/01/2016 – 08/31/2016	17,856	5 minutes	3.04M
(Li et al., 2025)	UrbanEV	275	09/01/2022 – 02/28/2023	4344	1 hour	1.19M
(Wang et al., 2020)	Know-Air	184	01/01/2015 – 12/31/2018	11688	3 hours	2.15M
(Li et al., 2018)	METR-LA	207	03/01/2012 – 06/27/2012	34,272	5 minutes	7.09M
LargeST (Liu et al., 2023b)	CA	8,600	01/01/2019 – 12/31/2019	35,040	15 minutes	30.13M
	GLA	3,834	01/01/2019 – 12/31/2019	35,040	15 minutes	13.43M
	GBA	2,352	01/01/2019 – 12/31/2019	35,040	15 minutes	8.87M
	SD	716	01/01/2019 – 12/31/2020	70,080	15 minutes	5.02M
(Chen and Liang, 2025)	Air-Stream	1087 → 1154 → 1193 → 1202	01/01/2016 - 12/31/2019	34065	1 hour	15.79M
		655 → 715 → 786				
	PEMS-Stream	→ 822 → 834 → 850 → 871	07/10/2011 - 09/08/2017	61,992	5 minutes	34.30M
	Energy-Stream	103 → 113 → 122 → 134	Unknown (245 days)	34,560	10 minutes	1.63M

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad \text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right|$$

• Baseline

- Classical Learning Backbones for STF
 - STAEformer / STTN / GWNet / STGCN / STID / ST-Norm
- Popular test-time training methods
 - TTT-MAE / TENT
- Classical online time series forecasting methods
 - OnlineTCN / FSNet / OneNet
- Advanced STF methods using test information.
 - CompFormer / DOST
- Efficient Learning Methods for Large-Scale STF
 - PatchSTG
- OOD Learning Methods for STF
 - STONE
- Continual Learning Methods for STF
 - EAC / STKEC



Effectiveness Study (RQ1)

- RQ1 (Effectiveness)** : Can ST-TTC have a consistent improvement on various types of models and datasets? Can ST-TTC outperform previous learning methods that leverage test data?

Table 2 Performance comparison of different models w/ and w/o ST-TTC on common benchmarks.

Models		Transformer-based						Graph-based						MLP-based					
		STAEformer (Lin et al., 2023a)			STTN (Xu et al., 2020)			GWNNet (Wu et al., 2019)			STGCN (Yu et al., 2018)			STID (Shao et al., 2022b)			ST-Norm (Deng et al., 2021)		
w/ ST-TTC		✗	✓	Δ(%)	✗	✓	Δ(%)	✗	✓	Δ(%)	✗	✓	Δ(%)	✗	✓	Δ(%)	✗	✓	Δ(%)
PEMS-03	MAE	17.00 $\pm$ 0.16	16.75 $\pm$ 0.14	↓ 1.47	18.12 $\pm$ 0.53	17.88 $\pm$ 0.50	↓ 1.32	16.73 $\pm$ 0.41	16.42 $\pm$ 0.19	↓ 1.85	18.41 $\pm$ 0.35	18.08 $\pm$ 0.35	↓ 1.79	17.48 $\pm$ 0.02	17.29 $\pm$ 0.01	↓ 1.09	17.27 $\pm$ 0.13	17.03 $\pm$ 0.12	↓ 1.39
	RMSE	29.98 $\pm$ 0.59	29.48 $\pm$ 0.58	↓ 1.67	31.02 $\pm$ 1.55	30.48 $\pm$ 1.37	↓ 1.74	28.48 $\pm$ 0.48	27.90 $\pm$ 0.13	↓ 2.04	31.74 $\pm$ 0.94	31.10 $\pm$ 0.86	↓ 2.02	29.10 $\pm$ 0.22	28.74 $\pm$ 0.21	↓ 1.24	29.28 $\pm$ 0.20	28.71 $\pm$ 0.14	↓ 1.95
	MAPE(%)	15.82 $\pm$ 0.22	15.82 $\pm$ 0.20	↓ 0.00	18.43 $\pm$ 1.19	18.01 $\pm$ 1.06	↓ 2.28	16.70 $\pm$ 0.60	16.49 $\pm$ 0.30	↓ 1.26	18.90 $\pm$ 0.78	18.68 $\pm$ 0.44	↓ 1.16	17.50 $\pm$ 0.12	17.39 $\pm$ 0.01	↓ 0.63	17.20 $\pm$ 0.82	16.92 $\pm$ 0.52	↓ 1.63
PEMS-04	MAE	19.48 $\pm$ 0.05	19.33 $\pm$ 0.06	↓ 0.77	20.63 $\pm$ 0.03	20.48 $\pm$ 0.03	↓ 0.73	20.57 $\pm$ 0.37	20.49 $\pm$ 0.39	↓ 0.39	20.70 $\pm$ 0.14	20.57 $\pm$ 0.12	↓ 0.63	19.97 $\pm$ 0.07	19.86 $\pm$ 0.08	↓ 0.55	20.22 $\pm$ 0.10	20.08 $\pm$ 0.09	↓ 0.69
	RMSE	32.58 $\pm$ 0.39	32.31 $\pm$ 0.34	↓ 0.83	33.14 $\pm$ 0.16	32.82 $\pm$ 0.10	↓ 0.97	32.64 $\pm$ 0.35	32.49 $\pm$ 0.38	↓ 0.46	33.11 $\pm$ 0.22	32.80 $\pm$ 0.20	↓ 0.94	32.62 $\pm$ 0.08	32.49 $\pm$ 0.08	↓ 0.40	33.15 $\pm$ 0.27	32.73 $\pm$ 0.23	↓ 1.27
	MAPE(%)	12.42 $\pm$ 0.01	12.37 $\pm$ 0.09	↓ 0.40	14.74 $\pm$ 0.76	14.53 $\pm$ 0.41	↓ 1.42	14.44 $\pm$ 0.47	14.43 $\pm$ 0.32	↓ 0.07	14.15 $\pm$ 0.16	14.04 $\pm$ 0.15	↓ 0.78	12.78 $\pm$ 0.17	12.72 $\pm$ 0.06	↓ 0.47	13.72 $\pm$ 0.13	13.68 $\pm$ 0.24	↓ 0.29
PEMS-07	MAE	21.67 $\pm$ 0.16	21.41 $\pm$ 0.13	↓ 1.20	23.30 $\pm$ 0.81	23.08 $\pm$ 0.75	↓ 0.94	22.62 $\pm$ 0.27	22.46 $\pm$ 0.27	↓ 0.71	24.26 $\pm$ 0.23	23.83 $\pm$ 0.18	↓ 1.77	21.72 $\pm$ 0.05	21.55 $\pm$ 0.05	↓ 0.78	22.69 $\pm$ 0.15	22.50 $\pm$ 0.14	↓ 0.84
	RMSE	37.48 $\pm$ 0.52	37.03 $\pm$ 0.48	↓ 1.20	37.55 $\pm$ 0.91	37.24 $\pm$ 0.81	↓ 0.83	36.87 $\pm$ 0.23	36.62 $\pm$ 0.26	↓ 0.68	39.31 $\pm$ 0.26	38.63 $\pm$ 0.20	↓ 1.73	36.24 $\pm$ 0.06	36.00 $\pm$ 0.06	↓ 0.66	38.14 $\pm$ 0.44	37.77 $\pm$ 0.41	↓ 0.97
	MAPE(%)	8.92 $\pm$ 0.05	8.87 $\pm$ 0.06	↓ 0.56	10.08 $\pm$ 0.24	9.98 $\pm$ 0.24	↓ 0.99	9.79 $\pm$ 0.16	9.75 $\pm$ 0.11	↓ 0.41	10.57 $\pm$ 0.18	10.38 $\pm$ 0.07	↓ 1.80	9.05 $\pm$ 0.03	9.00 $\pm$ 0.03	↓ 0.55	9.94 $\pm$ 0.56	9.86 $\pm$ 0.33	↓ 0.80
PEMS-08	MAE	14.84 $\pm$ 0.09	14.73 $\pm$ 0.08	↓ 0.74	17.19 $\pm$ 0.17	17.07 $\pm$ 0.16	↓ 0.70	16.37 $\pm$ 0.21	16.28 $\pm$ 0.20	↓ 0.55	17.33 $\pm$ 0.19	17.17 $\pm$ 0.21	↓ 0.92	15.61 $\pm$ 0.02	15.52 $\pm$ 0.02	↓ 0.58	16.69 $\pm$ 0.06	16.56 $\pm$ 0.04	↓ 0.78
	RMSE	25.61 $\pm$ 0.17	25.49 $\pm$ 0.16	↓ 0.47	27.07 $\pm$ 0.30	26.93 $\pm$ 0.29	↓ 0.52	26.14 $\pm$ 0.23	26.05 $\pm$ 0.21	↓ 0.34	27.49 $\pm$ 0.21	27.31 $\pm$ 0.22	↓ 0.65	25.70 $\pm$ 0.05	25.60 $\pm$ 0.05	↓ 0.39	26.94 $\pm$ 0.10	26.80 $\pm$ 0.10	↓ 0.52
	MAPE(%)	9.37 $\pm$ 0.04	9.32 $\pm$ 0.06	↓ 0.53	11.27 $\pm$ 0.26	11.20 $\pm$ 0.28	↓ 0.62	10.95 $\pm$ 0.34	10.79 $\pm$ 0.18	↓ 1.46	11.63 $\pm$ 0.36	11.53 $\pm$ 0.30	↓ 0.86	9.82 $\pm$ 0.12	9.76 $\pm$ 0.06	↓ 0.61	11.46 $\pm$ 0.89	11.19 $\pm$ 0.38	↓ 2.36
KnowAir	MAE	17.13 $\pm$ 0.19	17.06 $\pm$ 0.18	↓ 0.41	17.14 $\pm$ 0.16	17.06 $\pm$ 0.16	↓ 0.47	17.03 $\pm$ 0.07	16.94 $\pm$ 0.07	↓ 0.53	17.03 $\pm$ 0.06	16.96 $\pm$ 0.06	↓ 0.41	18.07 $\pm$ 0.11	17.98 $\pm$ 0.10	↓ 0.50	17.07 $\pm$ 0.01	17.01 $\pm$ 0.02	↓ 0.35
	RMSE	26.13 $\pm$ 0.20	26.06 $\pm$ 0.19	↓ 0.27	26.18 $\pm$ 0.16	26.13 $\pm$ 0.16	↓ 0.19	26.12 $\pm$ 0.15	26.05 $\pm$ 0.14	↓ 0.27	26.14 $\pm$ 0.17	26.07 $\pm$ 0.17	↓ 0.27	27.23 $\pm$ 0.02	27.17 $\pm$ 0.02	↓ 0.22	26.45 $\pm$ 0.07	26.39 $\pm$ 0.06	↓ 0.23
	MAPE(%)	62.14 $\pm$ 1.62	61.66 $\pm$ 1.53	↓ 0.77	64.12 $\pm$ 1.24	63.15 $\pm$ 1.40	↓ 1.51	64.51 $\pm$ 0.20	63.62 $\pm$ 0.28	↓ 1.38	63.12 $\pm$ 0.97	62.70 $\pm$ 0.84	↓ 0.67	70.00 $\pm$ 1.44	69.07 $\pm$ 1.35	↓ 1.33	60.76 $\pm$ 0.34	60.50 $\pm$ 0.63	↓ 0.43
UrbanEV	MAE	2.87 $\pm$ 0.02	2.85 $\pm$ 0.02	↓ 0.70	3.04 $\pm$ 0.06	2.99 $\pm$ 0.06	↓ 1.64	2.89 $\pm$ 0.03	2.85 $\pm$ 0.03	↓ 1.38	3.29 $\pm$ 0.10	3.23 $\pm$ 0.10	↓ 1.82	2.83 $\pm$ 0.01	2.79 $\pm$ 0.01	↓ 1.41	3.09 $\pm$ 0.02	3.04 $\pm$ 0.02	↓ 1.62
	RMSE	5.00 $\pm$ 0.01	4.98 $\pm$ 0.02	↓ 0.40	5.09 $\pm$ 0.07	5.03 $\pm$ 0.07	↓ 1.18	4.87 $\pm$ 0.07	4.81 $\pm$ 0.06	↓ 1.23	5.63 $\pm$ 0.23	5.52 $\pm$ 0.22	↓ 1.95	4.74 $\pm$ 0.02	4.67 $\pm$ 0.02	↓ 1.48	5.31 $\pm$ 0.03	5.22 $\pm$ 0.02	↓ 1.69
	MAPE(%)	27.14 $\pm$ 0.46	26.75 $\pm$ 0.58	↓ 1.44	28.75 $\pm$ 0.08	28.22 $\pm$ 0.29	↓ 1.84	29.10 $\pm$ 0.33	28.47 $\pm$ 0.26	↓ 2.16	31.67 $\pm$ 0.97	31.45 $\pm$ 1.01	↓ 0.70	27.60 $\pm$ 0.62	27.15 $\pm$ 0.43	↓ 1.63	29.53 $\pm$ 0.57	29.26 $\pm$ 0.57	↓ 0.91

- Consistent Effectiveness**

Table 3 Performance comparison of the advanced method w/ ST-TTC on *METR-LA* benchmark. [†] means the results are statistically significant (t-test with p-value < 0.01).

Method	MAE	RMSE	w/o Training Set	w/o Modifying Backbone
<i>The training / validation / test set split used below is 70% / 10% / 20%.</i>				
<i>TTT-MAE</i> (Gandelsman et al., 2022)	3.47±0.03	7.43±0.05	✗	✓
<i>TENT</i> (Wang et al., 2021)	4.84±0.08	8.53±0.10	✓	✓
<i>CompFormer*</i> (Zhang et al., 2023c)	3.46±0.02	7.19±0.08	✗	✓
ST-TTC	3.46±0.01 [†]	7.21±0.01	✓	✓
<i>The training / validation / test set split used below is 20% / 5% / 75%.</i>				
<i>OnlineTCN*</i> (Zinkevich, 2003)	4.78±0.03	8.70±0.04	✓	✗
<i>FSNet*</i> (Pham et al., 2023)	5.79±0.24	11.06±0.24	✓	✗
<i>OneNet*</i> (Wen et al., 2023)	4.94±0.03	8.80±0.06	✓	✗
<i>DOST*</i> (Wang et al., 2024c)	4.38±0.02	8.26±0.03	✓	✗
ST-TTC	3.77±0.07 [†]	7.75±0.13 [†]	✓	✓

- Competitive Effectiveness**



Universality Study (RQ2)

- RQ2 (Universality)**: Can ST-TTC effective in various real-world scenarios, including few-shot learning, long-term forecasting, and large-scale forecasting?

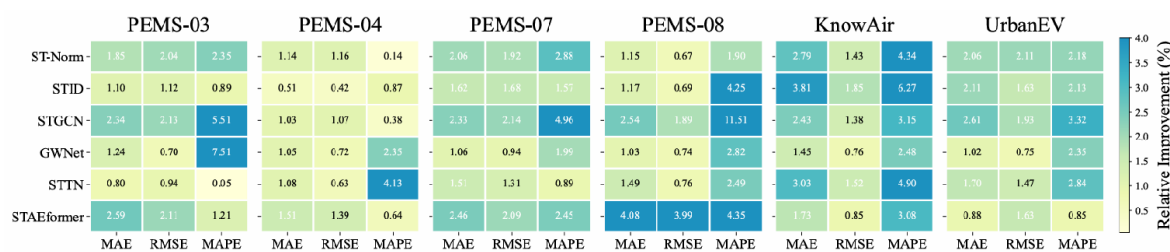


Figure 2 Relative improvements of different models w/ ST-TTC in the few-shot setting.

- Few-Shot Scenario**

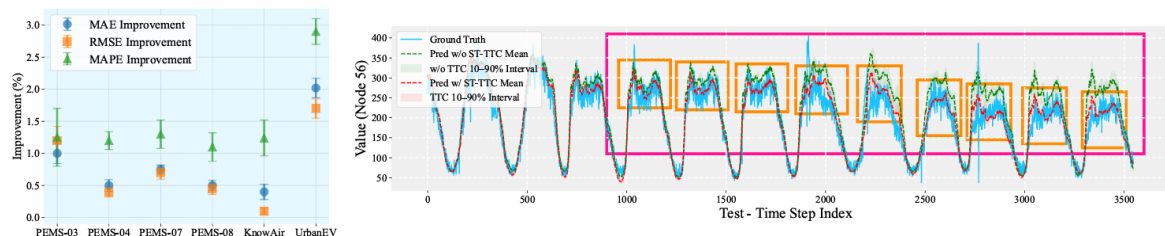


Figure 3 Left: relative improvement of long-term setting. Right: visualization study of PEMS-08.

- Long-Term Scenario**

Table E.2 PatchSTG with ST-TTC in LargeST Benchmark.

Datasets	Methods	Horizon 3			Horizon 6			Horizon 12			Average		
		MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
SD	PatchSTG	14.53	24.34	9.22	16.86	28.63	11.11	20.66	36.27	14.72	16.90	29.27	11.23
	w/ ST-TTC	14.44	24.07	8.70	16.66	28.18	10.93	20.37	35.68	14.27	16.72	28.83	11.11
	Δ	$\downarrow 0.09\%$	$\downarrow 1.1\%$	$\downarrow 5.6\%$	$\downarrow 1.2\%$	$\downarrow 1.6\%$	$\downarrow 1.6\%$	$\downarrow 1.4\%$	$\downarrow 1.6\%$	$\downarrow 3.1\%$	$\downarrow 1.1\%$	$\downarrow 1.5\%$	$\downarrow 1.1\%$
GBA	PatchSTG	16.81	28.71	12.25	19.68	33.09	14.51	23.49	39.23	18.93	19.50	33.16	14.64
	w/ ST-TTC	16.65	28.31	12.12	19.30	32.40	14.35	22.96	38.33	18.30	19.16	32.49	14.48
	Δ	$\downarrow 1.0\%$	$\downarrow 1.4\%$	$\downarrow 1.1\%$	$\downarrow 1.9\%$	$\downarrow 2.1\%$	$\downarrow 1.1\%$	$\downarrow 2.3\%$	$\downarrow 2.3\%$	$\downarrow 3.3\%$	$\downarrow 1.7\%$	$\downarrow 2.0\%$	$\downarrow 1.1\%$
GLA	PatchSTG	15.84	26.34	9.27	19.06	31.85	11.30	23.32	39.64	14.60	18.96	32.33	11.44
	w/ ST-TTC	15.78	26.08	9.15	18.76	31.29	11.21	22.86	38.89	14.35	18.69	31.78	11.36
	Δ	$\downarrow 0.4\%$	$\downarrow 1.0\%$	$\downarrow 1.3\%$	$\downarrow 1.6\%$	$\downarrow 1.8\%$	$\downarrow 0.8\%$	$\downarrow 2.0\%$	$\downarrow 1.9\%$	$\downarrow 1.7\%$	$\downarrow 1.4\%$	$\downarrow 1.7\%$	$\downarrow 0.7\%$
CA	PatchSTG	14.69	24.82	10.51	17.41	29.43	12.83	21.20	36.13	16.00	17.35	29.79	12.79
	w/ ST-TTC	14.59	24.61	10.40	17.14	28.97	12.51	20.76	35.38	15.54	17.10	29.31	12.53
	Δ	$\downarrow 0.7\%$	$\downarrow 0.8\%$	$\downarrow 1.0\%$	$\downarrow 1.6\%$	$\downarrow 1.6\%$	$\downarrow 2.5\%$	$\downarrow 2.1\%$	$\downarrow 2.1\%$	$\downarrow 2.9\%$	$\downarrow 1.4\%$	$\downarrow 1.6\%$	$\downarrow 2.0\%$

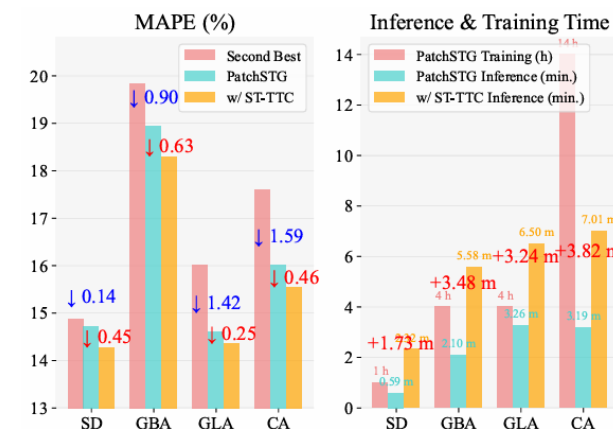


Figure 4 Performance on LargeST.

- Large-Scale Scenario**



Flexibility Study(RQ3)

- RQ3 (Flexibility)** : Can ST-TTC further enhance the performance of existing learning paradigms that utilize training data, such as OOD Learning and continual learning?

- OOD Learning Setting**

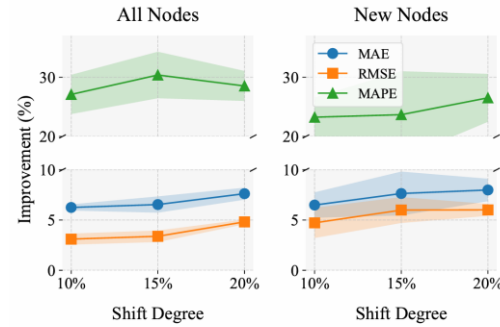


Figure 5 Relative improvement using our ST-TTC in the OOD learning setting.

- Continual Learning Setting**

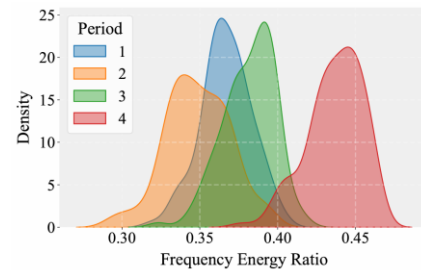


Figure 6 *Energy-Stream's* Shift.

Table E.3 Performance of spatio-temporal shift dataset SD-ratio(%) on all nodes and unknown new nodes at different spatio-temporal shift levels.

Dataset	Horizon	Methods	All Node			New Node		
			MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
SD-10%	12	STONE	40.74 $\pm$ 4.24	58.43 $\pm$ 3.67	45.82 $\pm$ 9.30	46.25 $\pm$ 7.02	66.97 $\pm$ 7.05	47.57 $\pm$ 8.02
		w/ ST-TTC	39.41 $\pm$ 3.82	57.44 $\pm$ 3.67	38.02 $\pm$ 5.19	44.65 $\pm$ 6.69	65.06 $\pm$ 6.95	41.85 $\pm$ 5.87
	Avg.	STONE	30.18 $\pm$ 1.19	42.83 $\pm$ 1.38	39.44 $\pm$ 3.03	32.79 $\pm$ 2.77	46.68 $\pm$ 3.87	40.12 $\pm$ 0.30
		w/ ST-TTC	28.29 $\pm$ 1.04	41.50 $\pm$ 1.16	28.66 $\pm$ 1.32	30.66 $\pm$ 2.63	44.44 $\pm$ 3.32	30.80 $\pm$ 1.54
SD-15%	12	STONE	35.31 $\pm$ 0.46	52.87 $\pm$ 0.57	34.05 $\pm$ 1.73	43.65 $\pm$ 0.85	66.07 $\pm$ 1.28	30.47 $\pm$ 1.40
		w/ ST-TTC	34.86 $\pm$ 0.15	52.25 $\pm$ 0.54	29.80 $\pm$ 0.79	41.93 $\pm$ 0.62	63.17 $\pm$ 0.29	27.70 $\pm$ 0.03
	Avg.	STONE	28.45 $\pm$ 0.05	41.30 $\pm$ 0.26	36.02 $\pm$ 2.91	33.20 $\pm$ 0.69	49.19 $\pm$ 0.97	29.77 $\pm$ 3.00
		w/ ST-TTC	26.59 $\pm$ 0.22	39.90 $\pm$ 0.09	24.96 $\pm$ 1.00	30.65 $\pm$ 0.29	46.24 $\pm$ 0.52	22.50 $\pm$ 0.38
SD-20%	12	STONE	36.13 $\pm$ 0.76	53.87 $\pm$ 0.44	34.19 $\pm$ 1.08	41.64 $\pm$ 1.23	63.19 $\pm$ 2.11	37.38 $\pm$ 3.29
		w/ ST-TTC	35.08 $\pm$ 1.08	52.37 $\pm$ 0.67	30.55 $\pm$ 1.13	40.10 $\pm$ 1.41	60.77 $\pm$ 2.07	33.70 $\pm$ 2.73
	Avg.	STONE	28.86 $\pm$ 0.13	41.72 $\pm$ 0.14	34.89 $\pm$ 1.15	31.46 $\pm$ 0.95	46.06 $\pm$ 1.60	36.35 $\pm$ 4.23
		w/ ST-TTC	26.67 $\pm$ 0.29	39.71 $\pm$ 0.08	24.92 $\pm$ 0.81	28.94 $\pm$ 0.79	43.29 $\pm$ 1.32	26.60 $\pm$ 2.49

Table 4 Performance comparison in continual learning setting.

Methods w/ ST-TTC		Air-Stream			PEMS-Stream			Energy-Stream		
		MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
EAC	✗	24.15 $\pm$ 0.14	38.22 $\pm$ 0.31	31.79 $\pm$ 0.05	14.92 $\pm$ 0.11	24.17 $\pm$ 0.17	20.82 $\pm$ 0.16	5.15 $\pm$ 0.10	5.46 $\pm$ 0.09	50.55 $\pm$ 2.60
	✓	23.54 $\pm$ 0.15	37.51 $\pm$ 0.27	31.44 $\pm$ 0.10	14.71 $\pm$ 0.07	23.87 $\pm$ 0.12	20.53 $\pm$ 0.03	3.47 $\pm$ 0.01	3.94 $\pm$ 0.00	39.66 $\pm$ 0.41
	Δ	↓ 2.5%	↓ 1.9%	↓ 1.1%	↓ 1.4%	↓ 1.2%	↓ 1.4%	↓ 32.6%	↓ 27.8%	↓ 21.6%
STKEC	✗	25.44 $\pm$ 1.05	40.11 $\pm$ 1.13	33.30 $\pm$ 1.64	16.25 $\pm$ 0.04	26.73 $\pm$ 0.07	22.33 $\pm$ 0.16	5.41 $\pm$ 0.15	5.72 $\pm$ 0.10	52.40 $\pm$ 1.10
	✓	24.26 $\pm$ 0.05	39.02 $\pm$ 0.05	31.73 $\pm$ 0.04	16.05 $\pm$ 0.06	26.39 $\pm$ 0.11	21.88 $\pm$ 0.06	3.83 $\pm$ 0.09	4.28 $\pm$ 0.08	43.22 $\pm$ 0.90
	Δ	↓ 4.6%	↓ 2.7%	↓ 4.7%	↓ 1.2%	↓ 1.3%	↓ 2.0%	↓ 29.2%	↓ 25.2%	↓ 17.5%



Mechanism & Robustness Study (RQ4)

- RQ4 (Mechanism & Robustness)** : How does ST-TTC work? Which components or strategies are crucial? Are these components or strategies sensitive to parameters or design?

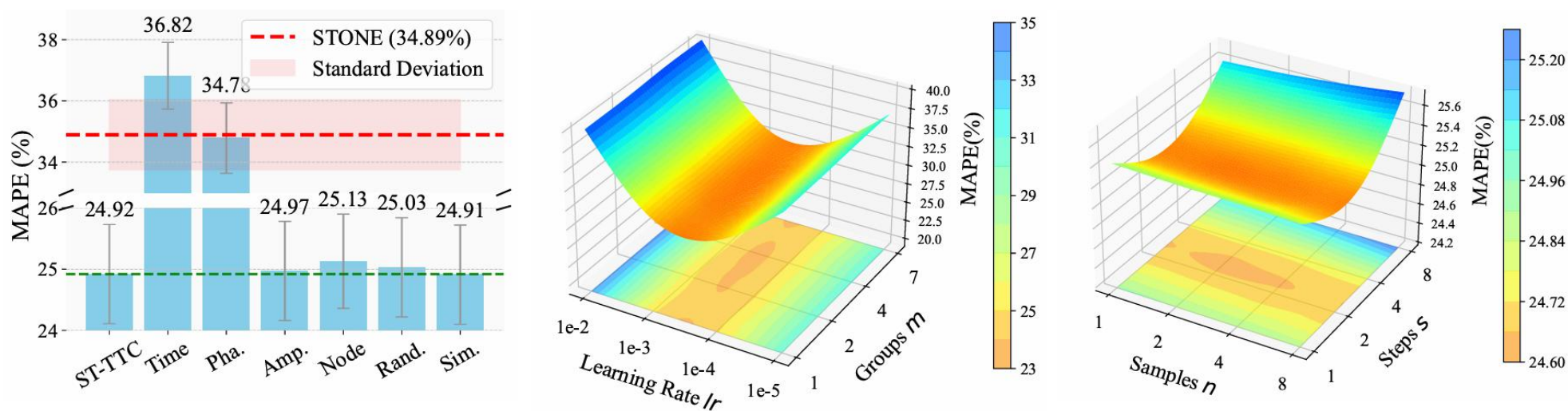


Figure 7 Left: Strategy comparison. Middle: Effect of lr w.r.t m . Right: Effect of n w.r.t s .

- Strategy Study**

- Parameter study**



Efficiency & Lightweight Study (RQ5)

- RQ5 (Efficiency & Lightweight)** : What is the time and parameter cost of ST-TTC during test-time computation, and how does it compare to other advanced methods?

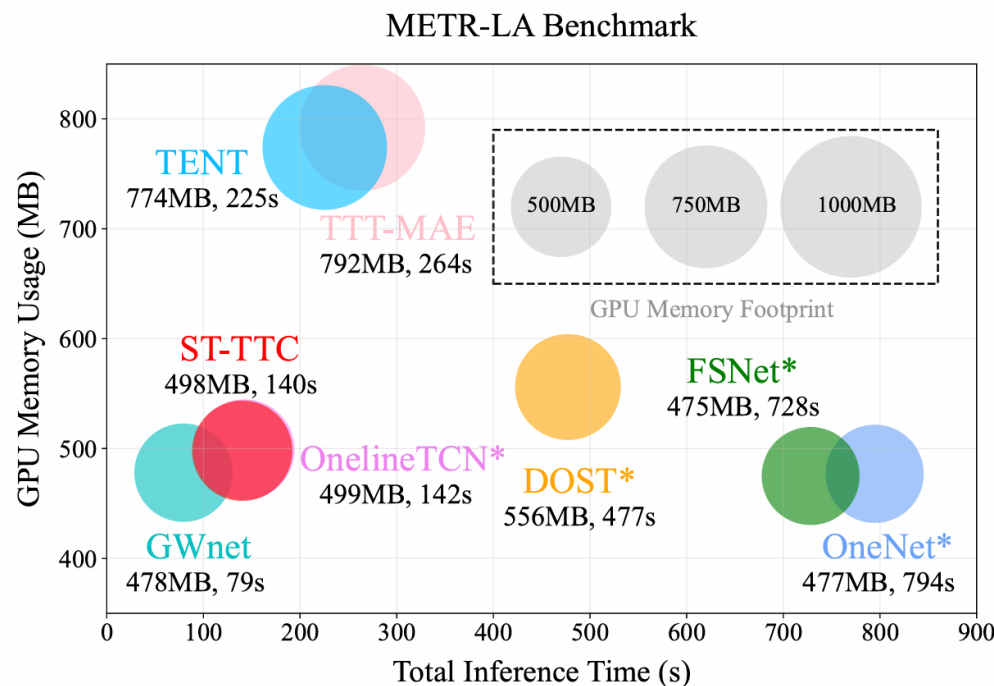


Figure 8 time and memory.



Discussion

- **Limitation**

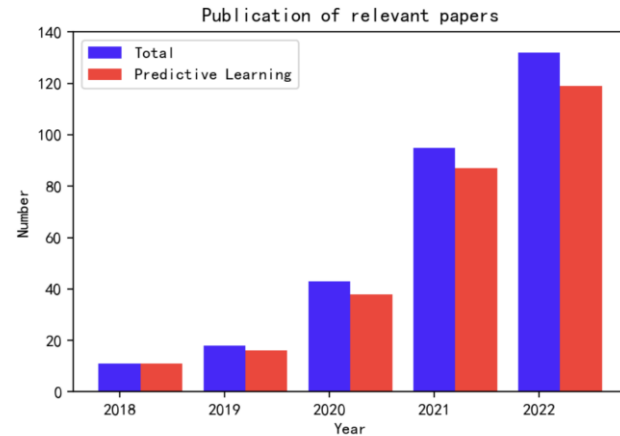
- ❶ Our current study does not involve testing on spatio-temporal foundation models. -> Reasonable but not comprehensive, we can still handle it.
- ❷ For commonly used small spatio-temporal benchmark datasets, the performance improvement is not yet significant. -> Research extending beyond this time span remains worthy of exploration, but retraining seems more appropriate in some context.
- ❸ We observed in our experiments that utilizing a larger amount of test information that is more similar to the current test sample does not significantly affect the results. -> Other technical, such as parameter sparsification and pruning, also merit investigation. But we provide a compress principle.

- **Future Work**

- ❶ Exploring how to integrate retrieval-augmented techniques to filter more effective learning samples from arbitrary external scenarios, thereby combining them with our test-time computation framework to optimize performance on small-scale datasets.. -> Large Foundation ST-Model



More discussion



- Graph Community:**

ChebNet (2016) → GCN(2017) → AGCN(2018)
→ GAT(2018) → GIN(2019)

- NLP Community:**

LSTM (1997) → TCN (2016) → Attention (2015)
→ Transformer (2017) → PLM (2020) → LLM (2022)

Table 1: Short-term Forecasting Performance.

Methods		STAEformer		STID		D ² STGNN		STGODE		STNorm		AGCRN		GWNET		ASTGCN		STGCN		DCRNN		LSTM	
Metric		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
PEMS03	3	13.96	23.56	13.81	<u>23.03</u>	13.76	23.81	14.82	24.48	14.24	24.01	14.46	24.91	<u>13.58</u>	23.20	15.31	25.43	15.49	26.80	14.21	24.32	14.55	24.70
	6	15.25	25.93	15.24	26.02	15.01	26.03	16.18	27.20	15.62	26.68	15.74	27.33	<u>14.75</u>	<u>25.39</u>	17.09	28.53	16.57	28.75	15.61	26.69	16.40	27.88
	12	17.66	29.85	17.41	29.41	17.03	29.14	18.41	31.08	17.26	29.32	17.31	29.85	<u>16.74</u>	<u>28.46</u>	20.64	34.34	18.41	31.65	17.88	30.16	19.82	32.94
	Avg	15.33	25.96	15.25	25.67	15.04	25.95	16.16	27.02	15.46	26.24	15.61	27.02	<u>14.81</u>	<u>25.31</u>	17.28	28.75	16.61	28.71	15.62	26.58	16.57	27.92
PEMS04	3	18.50	29.69	<u>17.73</u>	<u>28.73</u>	18.32	29.30	18.87	29.83	18.31	29.76	18.29	29.37	17.93	28.83	19.46	30.57	18.86	30.02	19.01	30.13	19.68	31.28
	6	19.74	31.49	<u>18.62</u>	<u>30.13</u>	19.38	30.87	19.92	31.50	19.06	31.38	19.04	30.57	18.93	30.33	20.85	32.62	19.54	31.29	20.72	32.57	21.88	34.68
	12	21.76	34.26	<u>20.06</u>	<u>32.18</u>	21.11	33.25	21.78	34.15	20.20	33.09	20.21	32.59	20.52	32.59	24.26	37.30	20.82	33.22	23.73	36.62	26.29	41.19
	Avg	19.74	31.44	<u>18.62</u>	<u>30.07</u>	19.37	30.79	19.93	31.46	19.00	31.11	19.02	30.66	18.92	30.29	21.12	32.92	19.57	31.26	20.82	32.64	22.17	35.06
PEMS07	3	19.88	32.01	<u>18.62</u>	<u>30.60</u>	19.19	31.05	20.24	32.08	19.21	31.48	19.12	31.22	18.93	30.84	21.91	33.84	20.97	33.29	20.26	32.27	20.43	33.03
	6	21.65	34.91	<u>19.96</u>	<u>33.00</u>	20.60	33.59	21.58	34.63	20.59	34.52	20.48	33.95	20.42	33.43	24.37	37.59	22.22	35.79	22.43	35.81	22.98	37.28
	12	24.66	39.39	<u>21.97</u>	<u>36.33</u>	22.88	37.34	23.94	38.48	22.61	38.25	22.33	37.31	22.81	37.04	29.23	44.64	24.29	39.41	26.32	41.50	27.76	44.51
	Avg	21.65	34.80	<u>19.91</u>	<u>32.83</u>	20.59	33.50	21.54	34.48	20.50	34.22	20.39	33.72	20.40	33.25	24.49	37.73	22.20	35.66	22.50	35.77	23.18	37.43
PEMS08	3	14.50	23.19	<u>13.44</u>	<u>21.63</u>	13.99	22.17	14.71	22.94	14.41	22.65	14.42	22.84	13.85	21.94	15.95	24.50	15.27	23.65	14.32	22.55	14.91	23.67
	6	15.80	25.50	<u>14.36</u>	<u>23.44</u>	14.92	23.92	15.62	24.73	15.30	24.65	15.34	24.91	14.79	23.70	17.54	26.96	15.99	25.23	15.42	24.56	16.39	26.53
	12	18.13	28.88	<u>15.80</u>	<u>25.80</u>	16.55	26.42	17.17	27.12	16.67	27.07	16.56	27.34	16.30	26.17	20.70	31.28	17.37	27.55	17.34	27.55	19.26	31.14
	Avg	15.85	25.41	<u>14.37</u>	<u>23.33</u>	14.95	23.83	15.59	24.55	15.26	24.47	15.30	24.80	14.80	23.62	17.64	26.96	16.03	25.18	15.46	24.50	16.52	26.57
PEMS-BAY	3	1.31	2.79	1.32	2.80	<u>1.29</u>	<u>2.71</u>	1.34	2.79	1.34	2.83	1.35	2.85	1.31	2.74	1.46	3.02	1.47	3.02	1.35	2.83	1.39	3.00
	6	<u>1.61</u>	3.67	1.65	3.73	<u>1.61</u>	<u>3.65</u>	1.67	3.72	1.66	3.80	1.67	3.81	1.64	3.70	1.90	4.07	1.80	3.93	1.70	3.85	1.83	4.19
	12	<u>1.88</u>	<u>4.33</u>	1.93	4.43	1.90	4.38	1.94	4.43	1.94	4.47	1.94	4.49	1.95	4.48	2.38	5.11	2.12	4.74	2.05	4.73	2.33	5.35
	Avg	<u>1.55</u>	<u>3.46</u>	1.58	3.52	<u>1.55</u>	<u>3.46</u>	1.60	3.52	1.59	3.57	1.61	3.59	1.58	3.51	1.84	3.91	1.75	3.77	1.64	3.67	1.78	4.01
METR-LA	3	2.80	5.47	2.82	5.55	<u>2.63</u>	<u>5.04</u>	2.83	5.49	2.80	5.51	2.87	5.59	2.70	5.18	3.04	5.71	2.82	5.48	2.81	5.35	2.97	5.86
	6	3.13	6.44	3.18	6.57	<u>2.96</u>	<u>6.02</u>	3.20	6.51	3.17	6.54	3.23	6.61	3.08	6.19	3.50	6.78	3.19	6.50	3.27	6.53	3.55	7.20
	12	3.49	7.34	3.57	7.54	<u>3.37</u>	<u>7.10</u>	3.59	7.48	3.56	7.50	3.59	7.55	3.50	7.25	4.09	8.00	3.61	7.54	3.91	7.94	4.36	8.82
	Avg	3.08	6.26	3.13	6.39	<u>2.93</u>	<u>5.90</u>	3.14	6.33	3.11	6.35	3.17	6.43	3.04	6.05	3.46	6.64	3.15	6.35	3.25	6.43	3.54	7.08

From DCRNN (2017) – STAEformer (2023), average ↓ MAE
in a **fair comparison** is **only ~1%**, what happened ?

From 2017-2023, it was party time of arc study for modeling ST Data Type

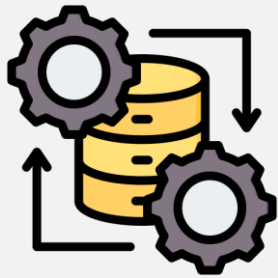
Now, the research area is over ? Yes, but not

There is still a lot of work to be explored in many stages,
including Data-Prepare / Pre-Train / Post-Train / Online-Deploy, etc.

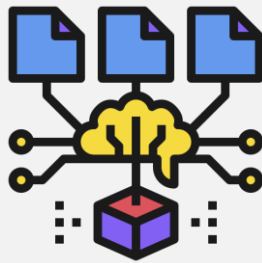
My part work about this research area

- How to use modern AI techniques (e.g., Machine Learning) to model spatio-temporal data in a *scalable and principled way* and move towards *general spatio-temporal intelligence*

All stages of AI modeling (i.e., Machine Learning) for ST data



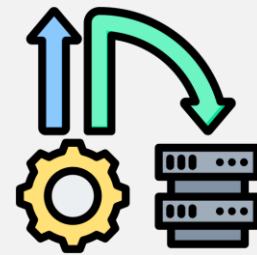
Data preparation



Pre-training



Post-training



Online deployment

Efficient

Versatile

Flexible

Dynamic

Modeling at four stages in a scalable and principled way

How about this research area in LLM era

From 2024-2025.05, I do some works in foundational machine learning methods for ST

Now, the research area is over ? Yes, but not

There is still a lot of work to be explored in LLM era (not just SFT / RL LLM w/ ST data, boring and terrible...).

The core view is how to see it as a physic world simulator and how to interaction w/ LLM.

But there's still a long way to go, so I think this research field is still in its very early stages...

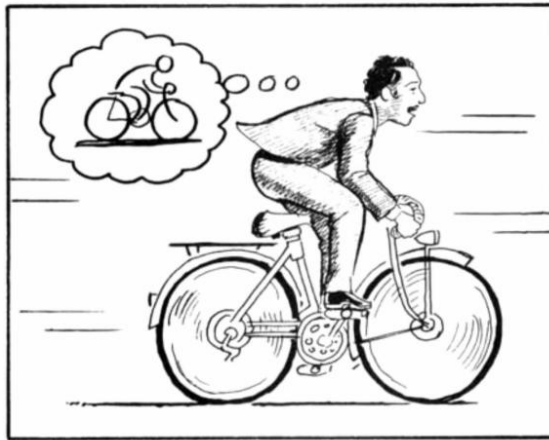


Figure 1. A World Model, from Scott McCloud's *Understanding Comics*. (McCloud, 1993; E, 2012)

$$P(s'|s) \rightarrow P(s'|s, \boxed{a})$$



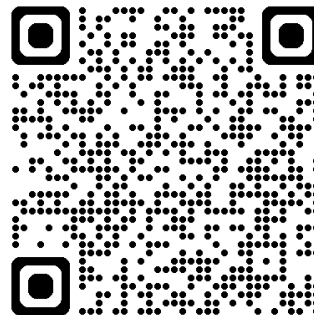
Thank you!

Q & A

WeChat



Code



Contact me: onedeanxxx@gmail.com

*The author is now more focused on
Everything related to Principled Scaling of LLM.*

Looking for collaboration opportunities or a suitable research internship / postdoc position!